DARSH PATEL

SOFTWARE ENGINEERING CO-OP CANDIDATE | MAY 2027 (12-16 MONTHS)

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EDUCATION

Western University | *London, ON* 2024 - Present

Bachelor of Engineering Science (B.E.Sc.) in Software Engineering

Relevant Coursework: Data Structures & Algorithms, Software Construction, Software Design, Database Systems, Computer Networks, Web Technologies, Discrete Structures, Digital Logic Systems, Quantum Computing

TECHNICAL SKILLS

Languages: Java, TypeScript, JavaScript, Python, C#, C++, SQL

Frontend: React, Next.js, Tailwind CSS, HTML, CSS, Vite, Java Swing

Backend & APIs: Spring Boot, Node.js, Express, Socket.IO, REST APIs, JWT/BCrypt, Flyway

Databases & Cloud: PostgreSQL, Neon, AWS (EC2, S3, IAM), Vercel, Docker

Computer Vision & APIs: OpenCV, MediaPipe, Gemini API, Twelve Data API

Developer Tools: Git, GitHub, Maven, Postman, VS Code, IntelliJ IDEA, Rider, Onshape

SELECTED PROJECTS

Personal Finance Tracker | *React, TypeScript, Spring Boot, PostgreSQL*

- Built and deployed a full-stack finance platform for transactions, monthly budgets, recurring expenses, investments, and analytics dashboards.
- Designed user-scoped REST APIs with JWT/BCrypt authentication, PostgreSQL persistence, and Flyway migrations; deployed using Vercel and Neon.
- Integrated Twelve Data market prices and Gemini financial insights with caching, timeouts, and notification triggers.

VisionControl | *Python, OpenCV, MediaPipe, Win32*

- Developed real-time dual-hand tracking with 21 landmarks per hand, EMA smoothing, calibrated gestures, pinch hysteresis, and swipe arbitration.
- Implemented gesture-driven cursor controls with configurable relative/absolute mapping; benchmarked ~28-31 FPS with two hands.
- Compared inference resolutions and applied landmark smoothing to balance responsiveness and recognition stability.

Argus | *TypeScript, Chrome MV3, Playwright | Team project*

- Co-developed an AI desktop agent; owned web-facing UI and browser integrations, including extension sensing and Playwright browsing tools.
- Built sensor-status displays, shortcuts, recording views, and terminal workflow UI; validated browser-to-agent interactions.

Bluff House | *React, TypeScript, Node.js, Socket.IO*

- Created a multiplayer Blackjack party game featuring private rooms, fictional chips, timed turns, and social bluff challenges.
- Developed real-time multiplayer state synchronization and responsive game interfaces using Socket.IO.

INDEPENDENT CONTRACT EXPERIENCE

Mechanical Design Contract | *Onshape CAD*

- Designed a custom ring bearing to support and stabilize a bench-mounted mechanical arm.
- Produced a complete Onshape CAD model, accounting for part fit, alignment, stability, and manufacturability.

HACKATHONS

DeenStudio - Western MSA Hackathon | *Collaborative product design and development*

- Co-designed the user experience and verse-to-video workflow for a Quran video creator featuring AI backgrounds, captions, and synchronized recitation.
- Collaborated on the overall implementation and prototype delivery; teammate performed a substantial portion of the direct coding.

Additional hackathon work: Argus (Nebius x NVIDIA Global AI Hackathon); Bluff House (Devpost Build With AI: Basics).